A Fiber-Net Criterion for BCP in Continuous $C_0(X)$-Algebras

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^*-algebra $C(K)$ has BCP/UBCP exactly when its pure state space, i.e. K, has a countable π-basis, the authors ask for a noncommutative version:

> $\{U_\alpha\}_{\alpha \in A}$ of K is called a π-basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].
>
> In the setting of commutative C^*-algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.
>
> **Corollary 3.9.** *Let $\mathcal{A}$ be a unital commutative C^*-algebra. Then the followings are equivalent:*
>
> (1) *$\mathcal{A}$ has the BCP;*
> (2) *the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π-basis;*
> (3) *$\mathcal{A}$ has the UBCP.*
>
> In particular, there exist commutative C^*-algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].
>
> We conclude this section by the following question.
>
> **Question 3.** *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^*-algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Let A be a continuous $C_0(X)$-algebra over a locally compact Hausdorff space X. We write A_x for the fiber at x, and $a(x) \in A_x$ for the image of $a \in A$. We use the standard fiber norm formula

$$\|a\| = \sup_{x \in X} \|a(x)\|.$$

Definition 1. *Let $0 < \delta < 1$. We say that A has a countable localizable δ-fiber-net if there is a countable family $\Sigma = \{\sigma_m : m \geq 1\}$ of bounded sections, $\sigma_m(x) \in A_x$, such that:*

1. $\sup_x \|\sigma_m(x)\| \leq 1$ *for every m;*
2. *for every $h \in C_c(X)$, the section $x \mapsto h(x)\sigma_m(x)$ belongs to A;*

3. *for every $a \in A$ and every m, the map*

$$(x, s) \mapsto \|a(x) - s\sigma_m(x)\|$$

is continuous on $X \times [0, 4]$;

4. *for every $x \in X$ and every $u \in S_{A_x}$, there is m such that $\|\sigma_m(x) - u\| < \delta$.*

This hypothesis is designed to include the trivial bundle $C_0(X, B)$ with separable nonzero fiber B: take constant bounded sections from a countable dense subset of S_B, then localize them by multiplying by functions in $C_c(X)$.

Theorem 1. *Let A be a continuous $C_0(X)$-algebra with $A_x \neq 0$ for every $x \in X$. Assume that A has a countable localizable $1/64$-fiber-net. Then the following are equivalent:*

1. *A has BCP;*
2. *A has UBCP;*
3. *X has a countable π-basis.*

Idea of the Proof

The previous continuous-field obstruction shows that BCP forces the base to have a countable π-basis. For the converse, the localizable fiber-net provides the missing noncommutative ingredient: a countable list of fiber directions. Near a point where $a \in S_A$ is large, choose a net section σ_m that approximates the normalized fiber value $a(x_0)/\|a(x_0)\|$. The center is then a localized section $4h_j\sigma_m$, where h_j is a scalar bump supported in a sufficiently small π-basic set. Convexity in the bump value $h_j(x) \in [0, 1]$ gives a fixed radius estimate.

Proof

We first recall the negative half, proved in the companion packet [5]. For convenience, we include the exact form needed here.

Lemma 1 (base obstruction)**.** *Let A be a continuous $C_0(X)$-algebra with $A_x \neq 0$ for all x. If A has BCP, then X has a countable π-basis.*

Proof. Given countably many candidate centers a_n, the sets

$$U_{n,k} = \{x : \|a_n(x)\| > \|a_n\| - 1/k\}$$

are open and nonempty whenever $a_n \neq 0$. If X has no countable π-basis, choose a nonempty open set W containing none of the nonempty $U_{n,k}$. Since all fibers are nonzero, choose $b \in A$ with $\|b\| = 1$ and $b(x) = 0$ outside W: take a with $a(x_0) \neq 0$ for some $x_0 \in W$, multiply by a scalar C_c-bump supported in W, and normalize. For each n, k, pick $x_{n,k} \in U_{n,k} \setminus W$. Then

$$\|b - a_n\| \geq \|a_n(x_{n,k})\| > \|a_n\| - 1/k,$$

and letting $k \to \infty$ gives $\|b - a_n\| \geq \|a_n\|$. Thus no countable family of balls $B(a_n, \|a_n\|)$ covers the unit sphere. □

It remains to prove that a countable π-basis gives UBCP under the fiber-net hypothesis. Fix a countable π-basis (V_j). By local compactness and regularity, choose for each j a function $h_j \in C_c(X)$ such that

$$0 \le h_j \le 1, \qquad \|h_j\|_\infty = 1, \qquad \operatorname{supp} h_j \subset V_j.$$

Let $\delta = 1/64$. For $j, m \ge 1$, put

$$c_{j,m} = 4h_j\sigma_m \in A.$$

Keep only those pairs (j, m) for which $\|c_{j,m}\| > 15/4$. We claim that the open balls

$$B(c_{j,m}, 7/2)$$

over this retained countable family cover S_A. Since each retained center has norm $> 15/4$, these balls are all disjoint from the open ball $B(0, 1/4)$. Thus this will prove UBCP.

Let $a \in S_A$. Choose $x_0 \in X$ with

$$\alpha := \|a(x_0)\| > 3/4.$$

Let $u = a(x_0)/\alpha \in S_{A_{x_0}}$. By the fiber-net property, choose m such that

$$\|\sigma_m(x_0) - u\| < \delta.$$

Write $w = \sigma_m(x_0)$. For $0 \le s \le 1$, the function

$$s \mapsto \|\alpha u - 4sw\|$$

is convex, hence bounded by the maximum of its endpoint values. At $s = 0$ the value is $\alpha < 1$. At $s = 1$,

$$\|\alpha u - 4w\| \le \alpha\|u - w\| + (4 - \alpha)\|w\| \le \alpha\delta + 4 - \alpha = 4 - \alpha(1 - \delta).$$

Since $\alpha > 3/4$ and $\delta = 1/64$, this is strictly less than $7/2$. Therefore

$$\sup_{0\le s\le 1} \|a(x_0) - 4s\sigma_m(x_0)\| < 7/2.$$

By the continuity assumption on σ_m, and compactness of $[0, 1]$, there is a neighborhood U of x_0 such that

$$\sup_{0\le s\le 1} \|a(x) - 4s\sigma_m(x)\| < 7/2 \qquad (x \in U).$$

Also $\|\sigma_m(x_0)\| > 1 - \delta$, so after shrinking U we may assume

$$\|\sigma_m(x)\| > 1 - 2\delta \qquad (x \in U).$$

Choose j with $V_j \subset U$.

For $x \in \operatorname{supp} h_j$, put $s = h_j(x)$. Since $\operatorname{supp} h_j \subset U$, the previous estimate gives

$$\|a(x) - c_{j,m}(x)\| = \|a(x) - 4h_j(x)\sigma_m(x)\| < 7/2.$$

For $x \notin \operatorname{supp} h_j$, we have $c_{j,m}(x) = 0$, so

$$\|a(x) - c_{j,m}(x)\| = \|a(x)\| \le 1 < 7/2.$$

Thus $\|a - c_{j,m}\| < 7/2$.

Finally, because h_j attains the value 1 at some point of $V_j \subset U$, the lower bound on $\|\sigma_m(x)\|$ on U gives

$$\|c_{j,m}\| \ge 4(1 - 2\delta) = 31/8 > 15/4.$$

Hence (j, m) is one of the retained pairs, and $a \in B(c_{j,m}, 7/2)$. This proves that A has UBCP.

Combining this UBCP implication with Lemma 1, and the trivial implication UBCP $\Rightarrow$ BCP, proves Theorem 1.

Scope and Limitations

The theorem is a partial answer to the source question, not a full characterization of arbitrary C^*-algebras. It applies to continuous fields whose fibers admit a countable family of localizable directions. It does not cover arbitrary upper-semicontinuous fields, nor C*-algebras with no useful central $C_0(X)$-structure. The point of the theorem is that it isolates a positive noncommutative replacement for the scalar bump directions used in the commutative proof.

Novelty and Search Bounds

The run indexes were searched for arXiv:2311.14257, C^*-algebras, BCP, UBCP, $C_0(X)$-algebras, continuous fields, fiberwise nets, and π-bases. Web searches on June 17, 2026 for continuous-field and $C_0(X)$-algebra BCP formulations found no exact existing version of this fiber-net criterion. The surrounding literature includes the source paper [1], the vector-valued $C_0(K, X)$ theorem [2], the Calkin obstruction [3], and the operator-space approximation criteria [4].

Human Review Focus

The main checks are:

- the definition of a localizable fiber-net, especially that it really includes constant bounded directions in $C_0(X, B)$;
- the compact-parameter continuity step for $(x, s) \mapsto \|a(x) - 4s\sigma_m(x)\|$;
- the constants 1/64, 7/2, and 15/4, which ensure both coverage and uniform separation from the origin.

References

[1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.

[2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.

[3] Qiyao Bao, Rui Liu, and Jie Shen, *The ball-covering property of non-commutative spaces of operators on Banach spaces*, arXiv:2412.07137, 2024.

[4] Qiyao Bao, Rui Liu, and Jie Shen, *Approximation properties and quantitative estimation for uniform ball-covering property of operator spaces*, arXiv:2507.02261, 2025.

[5] Run `fa_banach_001`, lane 19, *A Base-Space Obstruction for Continuous $C_0(X)$-Algebras with BCP*, solution packet `2311.14257_c0x_algebra_base_pi_obstruction`, 2026.